The Mag Estate

Plain Language Technical Summary

For AOS / AdventureNLearn · 2026-09-24

What this system is

A local AI operations platform running on a single Windows PC. No cloud dependency. Every component is locally hosted, locally verified, and locally recoverable.

1. Canon Verification

205 files in the canon. A SHA-256 manifest covers every file. A Python script recomputes all 205 hashes every morning. If any file changed, the audit reports which one and what the expected vs actual hash is.

```
import hashlib
from pathlib import Path
canon = Path('corpus/canon')
for line in (canon / 'MANIFEST.sha256').read_text().splitlines():
  parts = line.split(None, 1)
  want, rel = parts[0], parts[1].strip().lstrip('*')
  fp = canon / rel
  if fp.is_file():
    h = hashlib.sha256(fp.read_bytes()).hexdigest()
    if h != want: print(f'DRIFT: {rel}')
```

SHA-256 is NIST FIPS 180-4. The manifest pattern is content-addressable storage, identical to Git's internal object model.

2. Key Management

26 API keys across 14 providers, declared in one JSON manifest (configs/keys.json), loaded through one function (mag/key_law.py). The audit checks that every registered lane points at a key that actually exists.

Why: the NVIDIA lane was down for weeks because three files referenced the same key under three different names. No reader agreed on the name.

3. Multi-Lane Model Routing

10 API providers plus local Ollama. Each lane is measured for ok-rate, latency, tokens/second, and correctness.

Lane	OK	Correct	Latency	tok/s	Verdict
OpenRouter key 2	5/5	5/5	1.6s	114.9	carry
DeepSeek chat	5/5	5/5	1.1s	70.4	carry
OpenRouter key 1	5/5	5/5	3.8s	48.1	carry
NVIDIA glm-5.3-flash	5/5	5/5	25.6s	12.8	carry
Groq (both keys)	0/10	0/10	-	-	cut
Together	5/5	0/5	-	-	cut
Nous Hermes-4-70b	5/5	0/5	-	-	cut

Correct = the model cited 1976 for the Church Committee final report. Every failure has a named cause.

4. Concurrent Task Execution

A task wall queues real backlog chores and dispatches them across all lanes with ramping concurrency. Each lane starts at 1, ramps every 8 successful calls, backs off on 429/quota. 47 chores completed across 4 runs. Chores are real: label extraction, anchor notes, gap questions. Output lands in append-only JSONL ledgers.

5. Evidence Verification Pipeline

Six evidence families. Each verified by a two-pass process: a digger agent fetches primary sources and extracts verbatim quotes; an independent stranger agent re-fetches the URLs and checks every quote character-by-character.

Example - the Gilens & Page falsifier: downloaded the authors' replication file (4.4MB) from Cambridge Core, recomputed with pandas. N=1,779, overall adoption 33.05%, matching the paper. The popular '90% of the time elites win' claim appears nowhere in the paper. The 89.6% figure is the agreement share, not a capture rate.

6. Stereogram Decoding

A single-image autostereogram decoded using per-pixel shift correlation: for each 16x16 block, find which horizontal displacement produces the best match. The winning displacement per block forms a depth map. Result: a letter Q recovered from a 682KB image. Validated by round trip: encode a Q, decode a Q, same Q.

Pure NumPy + PIL. No ML model. No API call.

7. Scheduled Operations

Time	Task	What it does
09:00	Estate Spine Index	Walks every project file (1.2M files)
09:15	Estate Librarian	Shelf catalog + canon hash verification (205/205)
09:25	Estate Gopher	Label-farm chores on idle API keys
09:35	Estate FullScan	Semantic catalog: one-line brief per record
09:45	Estate KeyAudit	Key law audit: every lane resolves or fails

Each was fired manually at least once before being trusted to the scheduler.

What this system does NOT have

G7: a stranger-continuation test that passes without the original operator. Multi-day autonomous operation without human intervention. A formal succession plan tested by someone other than the operator. These are the same gaps.

Bibliography

1. Gilens & Page (2014). Perspectives on Politics, 12(3), 564-581. DOI 10.1017/S1537592714001595
2. Lermen et al. (2026). Large-scale deanonymization with LLMs. USENIX Security 2026. arXiv:2602.16800
3. Tyler & Clarke (1990). Autostereograms. Vision Research, 30(4).
4. NIST FIPS 180-4: Secure Hash Standard (SHA-256).
5. METR (2026). Time Horizons. metr.org/time-horizons/
6. IETF SCITT WG. draft-mih-scitt-agent-action-capsule-04. datatracker.ietf.org/
7. Senate Select Committee on Intelligence (1976). S. Rep. 94-755.
8. United States v. Epstein, S.D.N.Y. 1:19-cr-00490. CourtListener RECAP.

9. Doe v. United States, S.D. Fla. 9:08-cv-80736, Doc 435 (Marra, J., 2019).
10. Giuffre v. Maxwell, S.D.N.Y. 15 Civ. 7433, Doc 1315 (Preska, J., 2023).

Every claim above traces to a fetched URL, a run script, or a measured output. The metaphors are stripped. The numbers are the numbers.